Lecture 9

Vector fields and flows

Objectives. *A smooth vector field prescribes a velocity at every point of a manifold. Its integral curves describe the resulting motion. Comparing a second field along that motion gives the Lie bracket. We prove the connection between a zero bracket and commuting complete flows, then compute the rotation fields and their signs.*

By the end of this lecture you will be able to:

- Find integral curves and identify their maximal time domains.
- Compute a Lie bracket and derive it from backward transport by a flow.
- Prove the commuting-flow criterion for complete fields.
- Reconcile rotation-field and angular-momentum commutator signs.

9.1 Vector fields and related fields

Let M be a smooth n-manifold. A *smooth vector field* on M is a smooth map $X\colon M \to TM$ with $X(p) \in T_pM$. Such a map is called a *section* of the tangent bundle: composing it with the bundle projection gives the identity on M. In local coordinates it has the form $X = X^i\partial_i$, where the coefficient functions X^i are smooth. Indeed, its bundle-chart expression is $x \mapsto (x, X^1(x), \dots, X^n(x))$, so these two smoothness conditions are equivalent.

The field acts on smooth real functions by

$$Xf = X^i\partial_i f.$$

This is a smooth function. The product rule gives $X(fg) = (Xf)g + f(Xg)$, so X acts as a derivation of functions. The value $(Xf)(p)$ is the point derivation $X_p(f)$ of Lecture 8. The same definition applies to smooth functions on any open subset of M.

For a diffeomorphism $f\colon M \to N$, the *push-forward field* f_*X on N is given by

$$(f_*X)_q = \mathrm{d}f_{f^{-1}(q)}X_{f^{-1}(q)}.$$

The inverse selects exactly one starting point for each q. For a general smooth map f, fields X on M and Y on N are *f-related* when $\mathrm{d}f_pX_p = Y_{f(p)}$ for every $p \in M$. A field need not descend under a general map. For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ and $X = \partial_x$, the image vector is $2x\partial_y$ at $y = x^2$. The starting points 1 and -1 both map to 1 but give opposite nonzero vectors there. Thus no f-related field Y exists on the target. Lecture 10 compares this restriction with the pullback of forms.

9.2 Integral curves and maximal flows

An *integral curve* of X is a smooth curve $\gamma\colon I \to M$ on an open time interval satisfying $\gamma'(t) = X_{\gamma(t)}$. For initial point p, impose $0 \in I$ and $\gamma(0) = p$. In coordinates this is the

ordinary differential equation (ODE)

$$\dot{x}^i(t) = X^i(x(t)).$$

A curve is *maximal* if it has no extension to a larger open time interval as a solution with that initial condition. The resulting solution map is the *maximal flow*, written $\varphi\colon \mathcal{D} \to M$, with $\varphi_t(p) = \gamma_p(t)$. Its time domain at p is the interval $I_p = \{t : (t, p) \in \mathcal{D}\}$. A field is *complete* if $I_p = \mathbb{R}$ for every $p \in M$.

The *support* of a field is $\operatorname{supp} X = \overline{\{p : X_p \neq 0\}}$. The field has *compact support* when this closed set is compact.

Theorem 9.1 (Flow properties · *cited; laws and completeness proved from the cited flow properties*)**.** *A smooth vector field has a unique smooth maximal flow on an open set $\mathcal{D} \subseteq \mathbb{R} \times M$ containing $\{0\} \times M$. The time domains I_p are intervals. Its time maps satisfy*

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{s+t}(p) = \varphi_s(\varphi_t(p))$$

whenever both sides are defined. Each time map is a diffeomorphism between its open source and image, with inverse φ_{-t}. Every smooth field with compact support is complete.

Existence and uniqueness of maximal curves, openness of $\mathcal{D}$ and joint smoothness in (t, p) are cited from Lee, second edition, Theorem 9.12. Its local input is the ODE theorem in charts. We prove the laws and completeness from these properties.

Proof. First fix $t \in I_p$ and put $q = \varphi_t(p)$. The curve $u \mapsto \varphi_{t+u}(p)$ is an integral curve starting at q. Uniqueness identifies it with $u \mapsto \varphi_u(q)$ on their common interval. Maximality gives $I_q = I_p - t$: either larger interval would, by shifting back and using uniqueness, extend the other solution. This proves the group law and ensures that $\varphi_{-t}(q) = p$. Define $M_t = \{p \in M : t \in I_p\}$, open because $\mathcal{D}$ is open. Applying the inverse identity with $-t$ shows that $\varphi_t\colon M_t \to M_{-t}$ is onto and both inverse compositions are identities. Joint smoothness gives smooth time maps and inverses.

Next let $K = \operatorname{supp} X$ be compact. If K is empty, the field is zero and every solution is constant. Otherwise, openness of $\mathcal{D}$ gives, near each $p \in K$, an open set U_p and a time $\varepsilon_p > 0$ for which the flow is defined on $(-\varepsilon_p, \varepsilon_p) \times U_p$. Choose a finite subcover of K and let $\delta > 0$ be the minimum of these finitely many times. Solutions starting in K exist for at least time δ in both directions. Outside K, the constant curve is a solution and uniqueness makes it the maximal solution. Thus every starting point has a solution for $|t| < \delta$. If a maximal interval had a finite right endpoint b, restart the solution at a time $t < b$ with $b - t < \delta$. The new solution extends past b, contradicting maximality and uniqueness. The same argument at a finite left endpoint proves completeness. ∎

On $\mathbb{R}$, the field $X = x^2\partial_x$ has maximal flow

$$\varphi_t(x) = \frac{x}{1 - tx}, \qquad \mathcal{D} = \{(t, x) : 1 - tx > 0\}.$$

Differentiating gives $\partial_t \varphi_t = \varphi_t^2$, and the initial value is x. At $x = 0$ the solution is constant for all time. For $x > 0$ it diverges at $t = 1/x$. For $x < 0$ the divergence is at that negative time. These intervals are maximal because a continuous real-valued extension cannot pass through a divergence. Requiring only a nonzero denominator would include a separate branch beyond the pole. This field has noncompact support. Lecture 21 contrasts this example with smooth linear transport equations, whose solutions extend across each finite path interval. The graph shows the failure at $x = 1$.

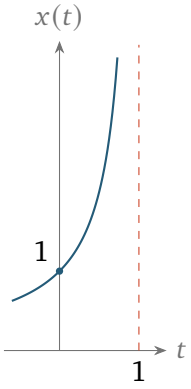


Figure 9.1: The solution starting at $x = 1$ diverges as t approaches 1 from below. The rational expression beyond the pole is not part of this integral curve.

9.3 The Lie bracket and matrix signs

For fields X and Y, define their *Lie bracket* by its action on smooth functions:

$$[X, Y]f = X(Yf) - Y(Xf).$$

The first composition expands as

$$X(Yf) = X^i(\partial_i Y^j)\partial_j f + X^i Y^j \partial_i \partial_j f.$$

In the second composition, swap the indices i and j in its second-derivative term. Equality of mixed partial derivatives makes those terms cancel. The result is

$$[X, Y] = \left(X^i \partial_i Y^j - Y^i \partial_i X^j\right)\partial_j.$$

The coefficients are smooth, so this expression is a smooth vector field. Its action on functions is intrinsic, making it independent of coordinates. Bilinearity, antisymmetry and Jacobi follow from the commutator calculation for operators, exactly as in Lecture 8.

The bracket preserves related fields under any smooth map f. Suppose X_j is f-related to Y_j for $j = 1, 2$. For every local smooth function h on the target, the chain rule gives $X_j(h \circ f) = (Y_j h) \circ f$. Applying it twice yields

$$[X_1, X_2](h \circ f) = ([Y_1, Y_2]h) \circ f.$$

Testing coordinate functions proves that the brackets are f-related. For a diffeomorphism this says $[f_* X, f_* Y] = f_*[X, Y]$. For a Lie group, write $\ell_g(h) = gh$ for left translation. A field is *left-invariant* when $(\ell_g)_* X = X$ for all g. Here a *matrix Lie group* means an embedded matrix subgroup with the smooth structure of Lecture 8. For tangent matrices, the formula $A^L(g) = gA$ follows by differentiating left multiplication.

Proposition 9.2 (Linear fields and left-invariant fields). *For $A, B \in \mathbb{R}^{n\times n}$, the fields $X_A(x) = Ax$ on $\mathbb{R}^n$ satisfy*

$$[X_A, X_B] = X_{-[A,B]}, \qquad [A, B] = AB - BA.$$

For tangent matrices A and B at the identity of a matrix Lie group, the left-invariant fields $A^L(g) = gA$ instead satisfy $[A^L, B^L] = [A, B]^L$.

Proof. For linear fields, $DX_A = A$ and $DX_B = B$. The coordinate formula gives $[X_A, X_B](x) = BAx - ABx$, proving the minus sign. The group fields are related by the inclusion i to the ambient fields $g \mapsto gA$ and $g \mapsto gB$. Their ambient derivatives in direction K are KA and KB. The related-fields result identifies their ambient bracket with the inclusion of the group bracket. Thus

$$\mathrm{d}i_g([A^L, B^L]_g) = (gA)B - (gB)A = g[A, B].$$

The inclusion has injective differential, so this identifies the group bracket. ∎

The operations act on different sides of their matrix arguments. They therefore give different signs when the coordinate bracket is computed.

For an abstract Lie group, each $A \in T_eG$ determines a left-invariant field by $A^L(g) = \mathrm{d}(\ell_g)_e A$. Smoothness follows by differentiating the smooth multiplication map in its second argument. Associativity and the chain rule give

$$(\ell_g)_* A^L(h) = \mathrm{d}(\ell_g)_h \mathrm{d}(\ell_h)_e A = \mathrm{d}(\ell_{gh})_e A = A^L(gh).$$

This proves invariance, and evaluation at the identity recovers A. Conversely invariance forces this formula, so the correspondence is a linear bijection. Naturality of the bracket makes $[A^L, B^L]$ left-invariant. Define $[A, B] = [A^L, B^L](e)$. This gives T_eG its Lie algebra structure. For matrix groups it agrees with the commutator just computed.

For example, the group of maps $r \mapsto ar+b$, where $a > 0$, has law $(a, b)(a', b') = (aa', b+ab')$. Set $H = \partial_a$ and $N = \partial_b$ at the identity $(1, 0)$. Differentiating left translation gives $H^L = a\partial_a$ and $N^L = a\partial_b$. The coordinate bracket gives $[H^L, N^L] = a\partial_b = N^L$.

The integral curve of A^L starting at the identity is a *one-parameter subgroup*: a smooth map $c\colon \mathbb{R} \to G$ with $c(s + t) = c(s)c(t)$ and $c(0) = e$. To prove this, take a local solution on $(-\delta, \delta)$ starting at the identity. Left translation gives a solution on that same interval starting at every group element. The uniform-time continuation argument above proves completeness. Left invariance and uniqueness identify the two curves $t \mapsto c(s + t)$ and $t \mapsto c(s)c(t)$, proving the group law. Conversely, differentiating a one-parameter subgroup's law makes it an integral curve of A^L, with $A = c'(0)$. For matrices the solution is $c(t) = e^{tA}$, where

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}.$$

If each $|A_{ij}| \le m$, multiplication of matrices gives $|(A^k)_{ij}| \le n^{k-1} m^k$ for $k \ge 1$, by induction. The resulting scalar exponential bounds the series and its differentiated series uniformly on bounded time intervals. Termwise differentiation gives $c'(t) = c(t)A$ and $c(0) = I$. The group-valued curve, viewed through the inclusion, solves this ambient equation. Ambient uniqueness identifies it with the series. The full field flow is $g \mapsto ge^{tA}$.

9.4 The Lie derivative of a field

Vectors at different points cannot be subtracted intrinsically. For $t \in I_p$, the map $\varphi_{-t}\colon M_{-t} \to M_t$ is a diffeomorphism and $\varphi_t(p) \in M_{-t}$. Thus it pushes $Y_{\varphi_t(p)}$ back to p by $\mathrm{d}\varphi_{-t}|_{\varphi_t(p)}$. The *Lie derivative* is the resulting derivative in T_pM:

$$(L_X Y)_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 ((\varphi_{-t})_* Y)_p.$$

For functions, define $L_X f = Xf$, the derivative of $f(\varphi_t(p))$ at zero.

Theorem 9.3 (Flow derivative and bracket · *proved given local flow existence and smooth dependence*). *The bracket $[X,Y]$ is a smooth vector field. The Lie derivative exists and satisfies $L_X Y = [X,Y]$.*

Proof. The bracket's coordinate formula above proves its smoothness. In a coordinate neighbourhood, the ODE gives $\varphi_t(p) = p + tX(p) + O(t^2)$. Differentiating the ODE $\partial_t \varphi_t = X(\varphi_t)$ in p, as joint smoothness permits, gives $\partial_t D\varphi_t = DX(\varphi_t) D\varphi_t$. Hence $D\varphi_t(p) = I + tDX_p + O(t^2)$. Inverting gives the inverse Jacobian $I - tDX_p + O(t^2)$ at $\varphi_t(p)$. Thus

$$\begin{aligned}((\varphi_{-t})_* Y)_p &= \big(I - tDX_p + O(t^2)\big)\big(Y_p + tDY_p X_p + O(t^2)\big) \\ &= Y_p + t\big(DY_p X_p - DX_p Y_p\big) + O(t^2).\end{aligned}$$

The coefficient of t is the bracket's coordinate formula. The definition by intrinsic push-forward makes this computation independent of the chosen neighbourhood. ∎

This is Tu, Theorem 20.4, with the same backward-push convention. Forward push would give the opposite sign.

9.5 Commuting complete flows

Suppose X and Y have flows φ_t and ψ_s. The fields *commute* when $[X,Y] = 0$. Their complete flows commute when $\varphi_t \circ \psi_s = \psi_s \circ \varphi_t$ for every $s, t \in \mathbb{R}$.

Theorem 9.4 (Commuting flows · *proved given flow existence, uniqueness and smooth dependence*). *For two complete smooth fields, their bracket vanishes if and only if their flows commute for all real times.*

Proof. The flow group law, the push-forward chain rule $(f \circ g)_* = f_* g_*$ and Theorem 9.3 give

$$\frac{\mathrm{d}}{\mathrm{d}t}(\varphi_{-t})_* Y = (\varphi_{-t})_* [X,Y].$$

Indeed, write the value at $t + h$ as $(\varphi_{-t})_*((\varphi_{-h})_* Y)$ and differentiate in h at zero. If the bracket is zero, this family is constant and equals Y. Hence $(\varphi_t)_* Y = Y$ for every t. For fixed t, both curves $s \mapsto \varphi_t(\psi_s(p))$ and $s \mapsto \psi_s(\varphi_t(p))$ solve the Y equation with the same initial point. Uniqueness makes them equal. Completeness gives the equality for all $s, t \in \mathbb{R}$.

Conversely, differentiate commuting flows in s at zero. The result is $\mathrm{d}\varphi_t|_p Y_p = Y_{\varphi_t(p)}$, so $(\varphi_{-t})_* Y = Y$. Differentiating in t at zero gives $[X,Y] = 0$. ∎

The coordinate fields ∂_x and ∂_y translate in independent directions, giving the rectangle in the figure. By contrast ∂_x and $x\partial_y$ have bracket ∂_y. Running their flows for times $u, v, -u, -v$ in that order ends at $(x, y + uv)$, rather than (x, y).

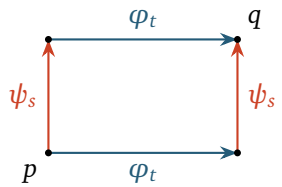


Figure 9.2: Translations in the two coordinate directions commute. The two orders reach the same opposite corner of the rectangle.

9.6 Distributions and Frobenius's theorem

A *smooth rank-k distribution* assigns a k-dimensional subspace $D_p \subseteq T_pM$ to every point, locally spanned by k independent smooth fields. It is *involutive* when the bracket of any two local fields taking values in D also takes values in D. An *integral manifold* is an immersion $i\colon S \to M$ satisfying

$$\mathrm{d}i_s(T_sS) = D_{i(s)} \qquad (s \in S).$$

> **Theorem 9.5** (Frobenius · *cited*)**.** *Every smooth involutive distribution of constant rank k has, near each point, coordinates in which it is spanned by $\partial_1, \dots, \partial_k$. Fixing the other coordinates gives local integral manifolds.*

This is Lee, second edition, Theorem 19.12. For example, the planes spanned by ∂_x and ∂_y in $\mathbb{R}^3$ have integral surfaces $z = c$. Involutivity checks consistency of prescribed tangent directions. The constant-rank hypothesis is part of the theorem.

9.7 Rotation fields and angular momentum

In Cartesian coordinates on $\mathbb{R}^3$, the rotation fields are

$$L_x = y\partial_z - z\partial_y, \qquad L_y = z\partial_x - x\partial_z, \qquad L_z = x\partial_y - y\partial_x.$$

For example, $L_x(x^2 + y^2 + z^2) = 2yz - 2zy = 0$. Cyclic permutation gives the same result for the other fields. Thus each is tangent to every sphere centred at zero. Its integral curves retain that radius. Here L_x names a field, distinct from the operator L_X acting on fields.

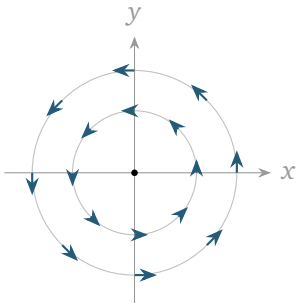


Figure 9.3: The field $L_z = (-y, x, 0)$ has circular trajectories in each horizontal plane. The arrows are its tangent velocities; the axis consists of fixed points.

Example 9.6 (Rotation flows and brackets)**.** The three flows are defined for all real t:

$$\begin{aligned}
\varphi^x_t(x, y, z) &= (x,\ y\cos t - z\sin t,\ y\sin t + z\cos t),\\
\varphi^y_t(x, y, z) &= (x\cos t + z\sin t,\ y,\ -x\sin t + z\cos t),\\
\varphi^z_t(x, y, z) &= (x\cos t - y\sin t,\ x\sin t + y\cos t,\ z).
\end{aligned}$$

For the first flow, differentiating its last two components gives $(-y\sin t - z\cos t,\ y\cos t - z\sin t)$, which is $(-z(t), y(t))$. At $t = 0$ the components are the initial coordinates. The same differentiation in the other coordinate planes verifies the other flows. Thus all three restrict to complete flows on S^2.

Their linear matrices are the basis matrices $A(1,0,0)$, $A(0,1,0)$ and $A(0,0,1)$ of Lecture 8. One bracket can also be read directly on coordinates: $[L_x, L_y]x = L_x(z) = y$, $[L_x, L_y]y =$

$-L_y(-z) = -x$ and $[L_x, L_y]z = 0$. Therefore

$$[L_x, L_y] = -L_z, \qquad [L_y, L_z] = -L_x, \qquad [L_z, L_x] = -L_y.$$

The matrices have the positive cyclic commutator, so this is precisely the linear-field minus sign of Proposition 9.2.

On complex smooth functions, define the angular-momentum differential operators $\widehat{L_a} = -\mathrm{i}\hbar L_a$ for $a \in \{x, y, z\}$, where $\hbar$ is the reduced Planck constant. Their operator commutator is

$$[\widehat{L_x}, \widehat{L_y}] = (-\mathrm{i}\hbar)^2(-L_z) = \hbar^2 L_z = \mathrm{i}\hbar\widehat{L_z}.$$

Thus the rotation-field sign agrees with the positive quantum angular-momentum relation. This is the differential-operator identification; no quantum representation theorem is assumed.

Self-check. Find the flow of $x\partial_y - y\partial_x$ on $\mathbb{R}^2$ and its value at time $\pi/2$ from $(1, 0)$. Include the trajectory from the origin.

Answer. The solution is $(x\cos t - y\sin t, x\sin t + y\cos t)$ for every $t \in \mathbb{R}$. From $(1, 0)$ it reaches $(0, 1)$ at time $\pi/2$, so positive time gives counter-clockwise rotation. The origin is a constant solution.

Summary. A vector field prescribes an ordinary differential equation in every chart. Uniqueness gives the flow group law, and compact support gives completeness. The Lie derivative of a field uses backward push-forward and equals its bracket with the generating field. Complete fields have zero bracket exactly when their flows commute. Linear fields Ax have the negative matrix commutator, whereas left-invariant matrix fields gA have the positive one. Lecture 10 introduces forms and their pullback under arbitrary smooth maps.

Reading. Tu, *An Introduction to Manifolds*, §14 and §20.2; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 9.12 and 19.12.